

# T3 UV one-loop RGE term comparison

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                           |
| $Y_d$                     | $-\bar{Q} Y_d H d + \text{h.c.}$                                   |
| $Y_e$                     | $-\bar{L} Y_e H e + \text{h.c.}$                                   |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$              |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$              |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$             |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$             |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$     |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$     |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$           |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$   |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2) (S_2^\dagger S_1)$      |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1) (S_1^\dagger T^A S_1)$  |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2) (S_2^\dagger T^A S_2)$  |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2 - 1)/4$                                                      |
| $T(d)$                    | $d(d^2 - 1)/12$                                                    |
| $C_{LLS_1 S_2}^{ij;AB}$   | $L_i L_j S_1^A S_2^B$                                              |
| $C_{ijab}$                | $L_i L_j \phi_a \phi_b$                                            |
| $C_5^{ij}$                | $(L_i^T C L_j) H H$                                                |

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use  $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$ . Gauge-coupling results use the same RGBeta  $g^2 \rightarrow g$  conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity:  $g_1$

$16\pi^2\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $\frac{59g_1^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{59g_1^3}{6}$ |

$\beta_{g_2}^{(1)}$   
Quantity:  $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{g_2^3}{2}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{g_2^3}{2}$ |

$\beta_{g_3}^{(1)}$   
Quantity:  $g_3$

$16\pi^2\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-7g_3^3$                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |

$\beta_{Y_u}^{(1)}$

Quantity:  $Y_u$

$$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $-3\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$ |
| T2     | $+ \text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$ |
| T3     | $+3\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   | T2                                                    | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $3\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$ | $+ \text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$ | $+3\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 2 of 3

| Column | Representative term structure                                                   |
|--------|---------------------------------------------------------------------------------|
| T1     | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$                                                 |
| T3     | $-8g_3^2\ (Y_u)_{ij}$                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                              | T2                              | T3                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$ | $-8g_3^2\ (Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 3 of 3

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

$\beta_{Y_d}^{(1)}$

**Quantity:**  $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$     |
| T2     | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ |
| T3     | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                  | T2                                                      | T3                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |

Quantity:  $Y_d$

Terms 2 of 3

| Column | Representative term structure                                                 |
|--------|-------------------------------------------------------------------------------|
| T1     | $+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_d)_{ij}$                                               |
| T3     | $-8g_3^2\ (Y_d)_{ij}$                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                            | T2                              | T3                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_d)_{ij}$ | $-8g_3^2\ (Y_d)_{ij}$ |

Quantity:  $Y_d$

Terms 3 of 3

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$

Quantity:  $Y_e$

$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

| Column | Representative term structure                             |
|--------|-----------------------------------------------------------|
| T1     | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$ |
| T2     | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$ |
| T3     | $+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                        | T2                                                        | T3                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$ | $+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$ |

Terms 2 of 2

| Column | Representative term structure                                                                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ |
| T2     | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$                                                                                                                                        |
| T3     | $-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$                                                                                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                | T2                                         | T3                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$ |

$\beta_{y_1}^{(1)}$

Quantity:  $y_1$

$$16\pi^2\,_{\mu}\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------|
| T1     | $T_{\nu}^{(1)}(y_1)_{ij}$                                                                                                      |
| T2     | $+\frac{1}{2}((y_2\ (y_2)^{\dagger}\ y_1)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_1)_{ij}+3\left(y_1\ (y_1)^{\dagger}\ y_1\right)_{ij})$ |
| T3     | $-\frac{9}{2}g_2^2\,(y_1)_{ij}$                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                        | T2                                                                                                                                                          | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $T_{\nu}^{(1)}(y_1)_{ij}$ | $+\frac{1}{2}((y_2\ (y_2)^{\dagger}\ y_1)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_1)_{ij}+3\left(y_1\ (y_1)^{\dagger}\ y_1\right)_{ij})-\frac{9}{2}g_2^2\,(y_1)_{ij}$ |    |

Quantity:  $y_1$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{3}{2}g_1^2(y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}g_1^2(y_1)_{ij}$ |

$\beta_{y_2}^{(1)}$   
Quantity:  $y_2$

$$16\pi^2\,{}_{\mu}\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $2T_{\nu}^{(2)}(y_2)_{ij}$                                       |
| T2     | $+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$ |
| T3     | $+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                         | T2                                                               | T3                                                               |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $2T_{\nu}^{(2)}(y_2)_{ij}$ | $+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$ | $+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$ |

Quantity:  $y_2$

Terms 2 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $+ \left( y_2 \, (y_2)^\dagger \, y_2 \right)_{ij}$ |
| T2     | $-\frac{9}{2} g_2^2 (y_2)_{ij}$                     |
| T3     | $-\frac{3}{2} g_1^2 (y_2)_{ij}$                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                  | T2                              | T3                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------|---------------------------------|---------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+ \left( y_2 \, (y_2)^\dagger \, y_2 \right)_{ij}$ | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ | $-\frac{3}{2} g_1^2 (y_2)_{ij}$ |

$\beta_{M_F}^{(1)}$

Quantity:  $M_F$

$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

| Column | Representative term structure                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------|
| T1     | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^{*}M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ |
| T2     | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$                                                                             |
| T3     | $-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$                                                                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                         | T2                                       | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^{*}M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$ |    |

$\beta_{m_1^2}^{(1)}$

Quantity:  $m_1^2$

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $m_1^2(2T_\nu^{(1)}+8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$     |
| T2     | $-4\operatorname{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$ |
| T3     | $-12g_2^2m_1^2$                                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                              | T2                                                                  | T3              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------------|---------------------------------------------------------------------|-----------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $m_1^2(2T_\nu^{(1)}+8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$ | $-4\operatorname{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$ | $-12g_2^2m_1^2$ |

Quantity:  $m_1^2$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-6g_1^2m_1^2$                |
| T2     | $+2\lambda_{12}^{(1)}m_2^2$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1             | T2                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------|-----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-6g_1^2m_1^2$ | $+2\lambda_{12}^{(1)}m_2^2$ |

$\beta_{m_2^2}^{(1)}$

Quantity:  $m_2^2$

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$$

| Column | Representative term structure                                   |
|--------|-----------------------------------------------------------------|
| T1     | $m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$                      |
| T2     | $-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$ |
| T3     | $+6\lambda_{12}^{(1)}m_1^2$                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                         | T2                                                              | T3                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------|-----------------------------------------------------------------|-----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$ | $-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$ | $+6\lambda_{12}^{(1)}m_1^2$ |

$\beta_{\lambda_1}^{(1)}$

Quantity:  $\lambda_1$

$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$

Terms 1 of 5

| Column | Representative term structure                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$                                                                          |
| T3     | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                        | T2                                                               | T3                                                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$ | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ |

Quantity:  $\lambda_1$

Terms 2 of 5

| Column | Representative term structure                                                         |
|--------|---------------------------------------------------------------------------------------|
| T1     | $-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$ |
| T2     | $+4\lambda_5\left(\lambda_5\right)^*$                                                 |
| T3     | $+ \frac{9g_2^4}{4}$                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                    | T2                                    | T3                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------------|---------------------------------------|----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$ | $+4\lambda_5\left(\lambda_5\right)^*$ | $+ \frac{9g_2^4}{4}$ |

Quantity:  $\lambda_1$

Terms 3 of 5

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ |
| T2     | $-9g_2^2\lambda_1$                               |
| T3     | $+\frac{3g_1^4}{4}$                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                               | T2                 | T3                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------|--------------------|---------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ | $-9g_2^2\lambda_1$ | $+\frac{3g_1^4}{4}$ |

Quantity:  $\lambda_1$

Terms 4 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+12\lambda_1^2$              |
| T2     | $+6\lambda_{HS_1}^{(1)2}$     |
| T3     | $+3\lambda_{HS_1}^{(A)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1               | T2                        | T3                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------|---------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+12\lambda_1^2$ | $+6\lambda_{HS_1}^{(1)2}$ | $+3\lambda_{HS_1}^{(A)2}$ |

Quantity:  $\lambda_1$

Terms 5 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+2\lambda_{HS_2}^{(1) \ 2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{HS_2}^{(1) \ 2}$ |

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity:  $\lambda_{S_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$     |
| T2     | $+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$                            |
| T3     | $-3\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                        | T2                                 | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------|------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$ | $+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$ | $-3\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $+21g_2^4$                                                              |
| T2     | $+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$ |
| T3     | $+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1         | T2                                                                      | T3                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------|-------------------------------------------------------------------------|--------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+21g_2^4$ | $+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$ | $+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+12g_1^4$                    |
| T2     | $+2\lambda_{12}^{(1)2}$       |
| T3     | $+4\lambda_{HS_1}^{(1)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1         | T2                      | T3                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------|-------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+12g_1^4$ | $+2\lambda_{12}^{(1)2}$ | $+4\lambda_{HS_1}^{(1)2}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure        |
|--------|--------------------------------------|
| T1     | $+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$ |
| T2     | $+14\lambda_{S_1}^{(1)^2}$           |
| T3     | $+26\lambda_{S_1}^{(A)^2}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                   | T2                         | T3                         |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------|----------------------------|----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$ | $+14\lambda_{S_1}^{(1)^2}$ | $+26\lambda_{S_1}^{(A)^2}$ |

$$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

Quantity:  $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                                   |
|--------|-----------------------------------------------------------------|
| T1     | $8\lambda_{S_2}^{(1)}T_\nu^{(2)}$                               |
| T2     | $-8\,\text{Tr}\Big(y_2\,(y_2)^\dagger\,y_2\,(y_2)^\dagger\Big)$ |
| T3     | $+6\lambda_{12}^{(1)2}$                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                | T2                                                              | T3                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------|-----------------------------------------------------------------|-------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $8\lambda_{S_2}^{(1)}T_\nu^{(2)}$ | $-8\,\text{Tr}\Big(y_2\,(y_2)^\dagger\,y_2\,(y_2)^\dagger\Big)$ | $+6\lambda_{12}^{(1)2}$ |

Quantity:  $\lambda_{S_2}^{(1)}$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)2}$     |
| T2     | $+10\lambda_{S_2}^{(1)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                        | T2                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)2}$ | $+10\lambda_{S_2}^{(1)2}$ |

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

**Quantity:**  $\lambda^{(1)}_{HS_1}$

$$16\pi^2\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu}=\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

### Terms 1 of 4

| Column | Representative term structure                                                                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda^{(A)}_{S_1})$ |
| T2     | $-2\operatorname{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$                                                                                                                                  |
| T3     | $+4\lambda_5\ (\lambda_5)^*$                                                                                                                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                     | T2                                                                    | T3                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda^{(A)}_{S_1})$ | $-2\operatorname{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+4\lambda_5\ (\lambda_5)^*$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $+6g_2^4$                                |
| T2     | $-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3     | $+3g_1^4$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1        | T2                                       | T3        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------|------------------------------------------|-----------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$ | $+3g_1^4$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$   |
| T2     | $+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T3     | $+6\lambda_1\lambda_{HS_1}^{(1)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                       | T2                                         | T3                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------|--------------------------------------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$ | $+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ | $+6\lambda_1\lambda_{HS_1}^{(1)}$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+4\lambda_{HS_1}^{(1)\ 2}$                 |
| T2     | $+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ |
| T3     | $+6\lambda_{HS_1}^{(A)\ 2}$                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                          | T2                                          | T3                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------|---------------------------------------------|-----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)\ 2}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{HS_1}^{(A)\ 2}$ |

$$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Quantity:  $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_2}^{(1)}(4T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                                                 |
| T3     | $+12\lambda_5\,(\lambda_5)^*$                                                                                                                                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                              | T2                                                              | T3                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{HS_2}^{(1)}(4T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_5\,(\lambda_5)^*$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 2 of 3

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$    |
| T2     | $-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$    |
| T3     | $+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                      | T2                                      | T3                                         |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------|-----------------------------------------|--------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$ | $-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$ | $+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 3 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+6\lambda_1\lambda_{HS_2}^{(1)}$           |
| T2     | $+4\lambda_{HS_2}^{(1)^2}$                  |
| T3     | $+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                | T2                         | T3                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------|----------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+6\lambda_1\lambda_{HS_2}^{(1)}$ | $+4\lambda_{HS_2}^{(1)^2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ |

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity:  $\lambda_{12}^{(1)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                                         |
|--------|---------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)}+4T_\nu^{(2)}+4\lambda_{S_1}^{(A)}\right)$       |
| T2     | $-4\,\text{Tr}\Big(y_1\,\left(y_1\right)^\dagger\,y_2\,\left(y_2\right)^\dagger\Big)$ |
| T3     | $+4\lambda_5\,\left(\lambda_5\right)^*$                                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                              | T2                                                                                    | T3                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)}+4T_\nu^{(2)}+4\lambda_{S_1}^{(A)}\right)$ | $-4\,\text{Tr}\Big(y_1\,\left(y_1\right)^\dagger\,y_2\,\left(y_2\right)^\dagger\Big)$ | $+4\lambda_5\,\left(\lambda_5\right)^*$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 2 of 3

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-12g_2^2\lambda_{12}^{(1)}$  |
| T2     | $-6g_1^2\lambda_{12}^{(1)}$   |
| T3     | $+4\lambda_{12}^{(1)2}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                           | T2                          | T3                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------|-----------------------------|-------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-12g_2^2\lambda_{12}^{(1)}$ | $-6g_1^2\lambda_{12}^{(1)}$ | $+4\lambda_{12}^{(1)2}$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$    |
| T2     | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$    |
| T3     | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                        | T2                                        | T3                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------|-------------------------------------------|----------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta_{\lambda_5}^{(1)}$   
Quantity:  $\lambda_5$

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)-4\lambda_{HS_1}^{(A)})$ |
| T2     | $-\frac{21g_2^2\lambda_5}{2}$                                                                                                                                                          |
| T3     | $-\frac{9g_1^2\lambda_5}{2}$                                                                                                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                     | T2                            | T3                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)-4\lambda_{HS_1}^{(A)})$ | $-\frac{21g_2^2\lambda_5}{2}$ | $-\frac{9g_1^2\lambda_5}{2}$ |

Quantity:  $\lambda_5$

Terms 2 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+2\lambda_{12}^{(1)}\lambda_5$   |
| T2     | $+2\lambda_1\lambda_5$            |
| T3     | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                              | T2                     | T3                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------|------------------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

Quantity:  $\lambda_5$

Terms 3 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

$$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

**Quantity:**  $\lambda^{(A)}_{HS_1}$

$$16\pi^2\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu}=\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda^{(A)}_{HS_1}\left(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda^{(A)}_{S_1}\right)$ |
| T2     | $-\frac{4}{3}\operatorname{Tr}\left(y_1\left(y_1\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$                                                                                                                                                    |
| T3     | $-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$                                                                                                                                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                     | T2                                                                                                  | T3                                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda^{(A)}_{HS_1}\left(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda^{(A)}_{S_1}\right)$ | $-\frac{4}{3}\operatorname{Tr}\left(y_1\left(y_1\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ | $-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ |
| T2     | $-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$                      |
| T3     | $+2\lambda_1\lambda_{HS_1}^{(A)}$                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                       | T3                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ | $-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ |
| T2     | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                           | T2                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity:  $\lambda_{S_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$$

Terms 1 of 4

| Column | Representative term structure                                                  |
|--------|--------------------------------------------------------------------------------|
| T1     | $\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$ |
| T2     | $+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$                                             |
| T3     | $-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                             | T2                                 | T3                                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$ | $+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$ | $-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 2 of 4

| Column | Representative term structure                                                               |
|--------|---------------------------------------------------------------------------------------------|
| T1     | $+ \frac{3 g_2^4}{4}$                                                                       |
| T2     | $+ g_1^2 \left( 9 g_2^2 - \frac{3 \lambda_{S_1}^{(1)}}{2} - 12 \lambda_{S_1}^{(A)} \right)$ |
| T3     | $+ g_2^2 \left( - 3 \lambda_{S_1}^{(1)} - 24 \lambda_{S_1}^{(A)} \right)$                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                    | T2                                                                                          | T3                                                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+ \frac{3 g_2^4}{4}$ | $+ g_1^2 \left( 9 g_2^2 - \frac{3 \lambda_{S_1}^{(1)}}{2} - 12 \lambda_{S_1}^{(A)} \right)$ | $+ g_2^2 \left( - 3 \lambda_{S_1}^{(1)} - 24 \lambda_{S_1}^{(A)} \right)$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 3 of 4

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+ \frac{3 g_1^4}{2}$               |
| T2     | $+ \frac{\lambda_{12}^{(1)2}}{4}$   |
| T3     | $+ \frac{\lambda_{HS_1}^{(1)2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                    | T2                                | T3                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------|-----------------------------------|-------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+ \frac{3 g_1^4}{2}$ | $+ \frac{\lambda_{12}^{(1)2}}{4}$ | $+ \frac{\lambda_{HS_1}^{(1)2}}{2}$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 4 of 4

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+3\lambda_{HS_1}^{(A)^2}$          |
| T2     | $+\frac{7\lambda_{S_1}^{(1)^2}}{4}$ |
| T3     | $+22\lambda_{S_1}^{(A)^2}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                         | T2                                  | T3                         |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------|-------------------------------------|----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)^2}$ | $+\frac{7\lambda_{S_1}^{(1)^2}}{4}$ | $+22\lambda_{S_1}^{(A)^2}$ |